

One gcd, four floors up

$$\gcd(2^a - 1, 2^b - 1) = 2^{\gcd(a,b)} - 1, \text{ read slowly}$$

Abstract

One fact about integers.

- `gcd_mersenne`: $\gcd(2^a - 1, 2^b - 1) = 2^{\gcd(a,b)} - 1$.

We watch it climb: *integers* $\rightarrow$ *a field has no exotic fixed points* $\rightarrow$ *Gold is APN* $\rightarrow$ *Kasami, even k , edge case*.

Each floor is short. Read each as *math* $\leftrightarrow$ *Lean* $\leftrightarrow$ *a typing judgement*.

The ground floor: a number

A **Mersenne number** is $2^a - 1$. The fact:

$$\gcd(2^a - 1, 2^b - 1) = 2^{\gcd(a,b)} - 1.$$

```
theorem gcd_mersenne (m n : ℕ) :  
  Nat.gcd (2 ^ m - 1) (2 ^ n - 1) = 2 ^ (Nat.gcd m n) - 1 :=  
  pow_sub_one_gcd_pow_sub_one 2 m n
```

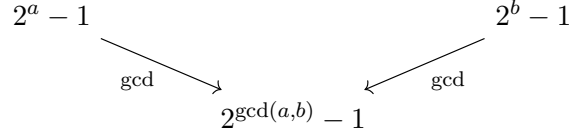
It is a *term* proof: the right-hand side *is* the proof. Mathlib already carries this under the name `pow_sub_one_gcd_pow_sub_one`. Our file only renames it and reads it slowly.

Why it is true (the picture)

Think of the cyclic group $\mathbb{Z}/(2^a - 1)$. The integer 2 acts by multiplication, and its order is exactly a (because $2^a \equiv 1$ and no smaller power does). So

$$2^a \equiv 1 \pmod{N} \iff (2^a - 1) \mid N.$$

Now $d \mid 2^a - 1$ and $d \mid 2^b - 1$ together say $2^a \equiv 2^b \equiv 1$ in the group of units mod d ; the smallest such exponent is $\gcd(a, b)$. Hence the common factors are exactly $2^{\gcd(a, b)} - 1$.



The map $a \mapsto 2^a - 1$ turns gcd of *exponents* into gcd of *Mersenne numbers*. It is a tiny “functor”: it sends the lattice $(\mathbb{N}, \gcd)$ into the lattice $(\mathbb{N}, \gcd)$, monotonically.

1 First floor: a field has no exotic fixed points

Move to $F = \text{GF}(2^n)$, a field of size 2^n . Ask: which x satisfy $x^{2^k} = x$? In char 2 this is the Frobenius fixed-point equation.

$$\Gamma = (F : \text{Type}, [\text{Field } F], [\text{Fintype } F], [\text{CharP } F \ 2])$$

```

lemma frob_fixed_implies_GF2 {F : Type*} [Field F] [Fintype F] [CharP F 2]
  {n : ℕ} (hn : Fintype.card F = 2 ^ n)
  {k : ℕ} (hk : 0 < k) (hcop : Nat.Coprime k n)
  {x : F} (hfixed : x ^ (2 ^ k) = x) :
  x = 0 ∨ x = 1

```

Where the gcd enters

Suppose $x \neq 0$. Two facts about its multiplicative order:

- from $x^{2^k} = x$ we get $x^{2^k - 1} = 1$;
- from finiteness (Fermat) we get $x^{2^n - 1} = 1$.

So the order of x divides *both* $2^k - 1$ and $2^n - 1$, hence divides their gcd:

$$x^{\gcd(2^k - 1, 2^n - 1)} = 1.$$

Now apply the ground floor:

$$\gcd(2^k - 1, 2^n - 1) = 2^{\gcd(k, n)} - 1 = 2^1 - 1 = 1 \quad (\text{since } \gcd(k, n) = 1).$$

Therefore $x^1 = 1$, i.e. $x = 1$. So $x \in \{0, 1\}$.

The hypothesis $\text{Nat.Coprime } k \ n$ ($\gcd(k, n) = 1$) is the *only* reason the answer is the prime subfield $\{0, 1\} = \mathbb{F}_2$ and nothing larger. Drop coprimality and the fixed field grows to $\text{GF}(2^{\gcd(k, n)})$.

$$\frac{x^{2^k-1} = 1 \quad x^{2^n-1} = 1 \quad \gcd(2^k - 1, 2^n - 1) \stackrel{\text{ground}}{=} 2^{\gcd(k, n)} - 1 = 1}{x^1 = 1} \text{ ORDER}$$

2 Second floor: Gold is APN

The **Gold** function is $x \mapsto x^{2^k+1}$. **APN** (almost perfect nonlinear) means: for every $a \neq 0$, the equation $f(x+a) + f(x) = b$ has at most 2 solutions.

```
theorem gold_is_apn {F : Type*} [Field F] [Fintype F] [CharP F 2]
  {n : ℕ} (hn : Fintype.card F = 2 ^ n)
  (k : ℕ) (hk : 0 < k) (hcop : Nat.Coprime k n) :
  IsAPN (fun x : F => x ^ (2 ^ k + 1))
```

The collision collapses to a fixed point

Expand the Gold difference (char 2, Freshman's Dream):

$$(x+a)^{2^k+1} + x^{2^k+1} = a^{2^k}x + ax^{2^k} + a^{2^k+1}.$$

A collision of x and y forces (after cancelling) the homogeneous equation

$$a^{2^k}(x+y) + a(x+y)^{2^k} = 0.$$

Divide by a^{2^k+1} and set $u = (x+y)/a$:

$$u^{2^k} = u.$$

That is the first-floor equation! By `frob_fixed_implies_GF2`, $u \in \{0, 1\}$, i.e. $y = x$ or $y = x + a$. At most two solutions. APN. ■

In Lean this is literally one call: `frob_fixed_implies_GF2 hn hk hcop` closes the kernel step inside `gold_is_apn`. The gcd fact is doing the load-bearing work two floors down.

$$\text{collision} \implies u^{2^k} = u \xRightarrow{\text{frob_fixed}} u \in \{0, 1\} \implies \text{APN}$$

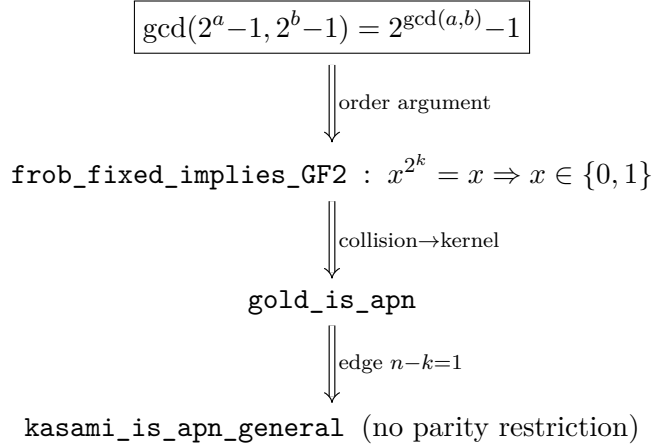
3 Third floor: Kasami, even k , at the edge

The Kasami exponent is $d_k = 2^{2k} - 2^k + 1$. The main theorem `kasami_is_apn` historically needed k **odd**. The even- k case is handled by a Frobenius twist that swaps $k \leftrightarrow n - k$. At the *edge* $n - k = 1$ the companion exponent is the Gold exponent $d_1 = 3 = 2^1 + 1$:

```
lemma kasami_one_is_apn {F : Type*} [Field F] [Fintype F] [CharP F 2]
  {n : ℕ} (hn : Fintype.card F = 2 ^ n)
  (hcop : Nat.Coprime 1 n) :
  IsAPN (fun x : F => x ^ kasamiExp 1) := by
  rw [kasamiExp_one] -- kasamiExp 1 = 3
  exact gold_is_apn hn 1 one_pos hcop
```

So the even- k Kasami argument *falls back on Gold* at its boundary, and Gold falls back on the fixed-point lemma, which falls back on the Mersenne gcd. Four floors, one staircase.

The staircase



One-screen summary

floor	statement	engine
ground	$\gcd(2^a-1, 2^b-1) = 2^{\gcd(a,b)}-1$	cyclic order of 2
1	$x^{2^k}=x \Rightarrow x \in \{0, 1\}$	order $\mid \gcd = 1$
2	Gold x^{2^k+1} is APN	collision $\rightarrow$ fixed point
3	Kasami even- k edge	companion = Gold ($d_1 = 3$)

```
gcd_mersenne  $\Rightarrow$  frob_fixed_implies_GF2  $\Rightarrow$  gold_is_apn  $\Rightarrow$ 
kasami_is_apn_general.
```